PROBLEM STATEMENT #45 | Developer Tools & IT Operations

Multi-Cloud Asset Cost Optimization Portal

1. Problem Context & Overview

A FinOps analytics portal ingesting AWS, Azure, and GCP billing CSVs, identifying unattached storage volumes and idle compute instances, and dispatching budget breach alerts.

Target Stakeholders / Actors: *FinOps Analyst, Cloud Administrator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall ingest cloud billing CSV exports, categorize costs by service and project tag, and highlight compute instances with < 5% CPU utilization over 14 days.
Sample Acceptance Criteria: Pass: Idle cloud asset flagged with monthly savings estimate; Fail: Active instance misclassified as idle.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.